MIL-L-52733A(ME)
1 July 1986
SUPERSEDING
MIL-L-52733(ME)
30 August 1972

MILITARY SPECIFICATION

LABORATORY, AIRMOBILE, AVIATION FUEL

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of airmobile aviation fuel laboratory, fully equipped with testing equipment and supplies, for testing aviation fuel.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.

MILITARY

MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-C-3774	- Crates, Wood; Open 12,000- and 16,000-Pound Capacity.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6640
DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:31Z
Check the source to verify that this is the current version before use.

MIL-B-7883	- Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-C-53039	- Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1474	- Noise Limits for Army Materiel.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

TA13219E1400 - Laboratory, Airmobile, Aviation Fuel.

SUPPLY CATALOG

SC-6640-97-CL-E01 - Laboratory, Airmobile, Aviation Fuel.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:31Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

B2.1 - Welding Procedure and Performance Qualification.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, NY 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

National Electrical Code (NEC)

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York, NY 10001.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The airmobile aviation fuel laboratory (hereinafter called "laboratory") shall be in accordance with TA13219E1400 and as specified herein. The laboratory shall be equipped with all items specified in SC-6640-97-CL-E01.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the (laboratory). Any data (e.g., shop drawings, layouts, flow

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:31Z
Check the source to verify that this is the current version before use.

sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.4).

3.3 Materials. Materials shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The laboratory shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the laboratory may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the laboratory may be newly fabricated from recovered materials to the maximum extent practicable, provided the laboratory produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the laboratory.

3.4 Fire extinguisher. Unless otherwise specified (see 6.2), the Government will furnish two fire extinguishers, vaporizing liquid, bromochlorodifluoromethane, 5-pound, NSN 4210-01-087-4989. The mountings shall be as specified in the applicable drawing (see 6.3).

3.5 Automatic gas alarm. The automatic gas alarm shall be calibrated and installed in accordance with the manufacturer's instructions.

3.6 Water system. The water system shall show no evidence of leakage when tested as specified in 4.5.2.3.

3.7 Air-vacuum system. The air-vacuum system shall show no evidence of leakage when tested as specified in 4.5.2.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:31Z
Check the source to verify that this is the current version before use.

3.8 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.9 Treatment and painting. Treatment and painting shall be in accordance with the applicable drawing. Unless otherwise specified (see 6.2), the laboratory shall then be overcoated in accordance with Government furnished camouflage patterns with MIL-C-46168 or MIL-C-53039 top coat of the colors specified in the camouflage patterns.

3.10 Wiring and electrical supplies. Wiring and electrical supplies shall conform to drawing requirements and shall conform under all conditions to the approved standards of the National Fire Protection Association's National Electrical Code (NEC).

3.11 Wiring protection. All wiring shall be firmly secured to prevent rubbing and abrasion. Suitable grommets shall be used at all points where wires which are not in conduit or loom, pass through body frame members, cabinet walls, or panels.

3.12 Wire identification. All wires shall be identified in such a manner as to facilitate circuit tracing and the establishment of failures.

3.13 Equipment location. Equipment furnished shall be located in the laboratory in accordance with the technical manual, TM 5-6040-213-14, for the laboratory (see 3.14).

3.14 Government-furnished property. When specified (see 6.2), the following property in the quantities indicated will be furnished by the Government (see 6.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:31Z
Check the source to verify that this is the current version before use.

Item No.	Description	Identification	Qty for/ea laboratory
1.	Shelter, electrical equip.	NSN 5411-01-092-0892	1
2.	Air conditioner, 9000 BTU/HR 208V, 50/60 Hz.	NSN 4120-00-926-1137	1
3.	Maintenance and repair procedure for shelters, electrical equipment.	TB-43-0124	1
4.	Air conditioner manual.	TM-5-4120-274-15	1
5.	Air conditioner manual.	TM-5-4120-274-25P	1
6.	Technical manual for laboratory, airmobile, aviation fuel.	TM-5-6640-213-14	1
7.	Pattern drawings.	Camouflage patterns	As required

3.15 Air transportability. All equipment, supplies, chemicals, binders, books, and manuals shall be secured within the laboratory to permit air transportability.

3.16 Noise limits. The noise produced by the laboratory shall conform with MIL-STD-1474 requirements when tested in accordance with 4.5.2.5. The noise limits for the interior of the laboratory shall conform to MIL-STD-1474, table 2, category E. The provisions of MIL-STD-1474, 4.3 and 4.4 shall be applicable to the exterior of the laboratory if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85 dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474, 4.3 and be readable at the distance indicated on the sign.

3.17 Safety. All rotating or reciprocating parts and other parts subject to high operational temperatures that are of such a nature or are so located as to be a hazard to operating personnel, shall be guarded or insulated to the extent necessary to eliminate the hazard. Protective devices shall not impair the operating functions.

3.18 Workmanship. There shall be no evidence of bulges, warpage, dirt fins, sprues, scale, flux, burrs, scratches, pits, sharp edges, rust, defective connections, loose, damaged, or missing parts.

3.18.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the metal may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated metal shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:31Z
Check the source to verify that this is the current version before use.

3.18.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the laboratory to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.1875 inch, unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the laboratory shall be at the option of the contractor.

3.18.3 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS B2.1 - Welding Procedure and Performance Qualification.
ASME - Boiler and Pressure Vessel Code, Section IX, Welding Qualification.

Certification shall be made available for review by the contracting officer or designated representative. Contractors who make only horizontal welds need not qualify welders for "all position welding".

3.18.4 Brazing. Brazing of steels, copper alloys and nickel alloys shall conform to MIL-B-7883.

3.18.5 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.18.6 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, conecentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

3.18.7 Machine work. Tolerances and gages for metal fits shall conform to the limits specified herein and on applicable drawing and to the standards of the industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:31Z
Check the source to verify that this is the current version before use.

all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Disassembly inspection. Failure of any test by the first article shall be cause for disassembly, in the presence of a Government representative, of the first article to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article laboratory shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article laboratory shall be tested as specified in 4.5.2.1 through 4.5.2.5. Failure of any test shall be cause for performance of the inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Examination. Each laboratoy shall be examined for defects marked "X" in column 2 of table I. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each laboratory shall be tested as specified in 4.5.2.1 through 4.5.2.5. Failure of any test shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:31Z
Check the source to verify that this is the current version before use.

4.5 Inspection schedule.

4.5.1 Examination. Examination shall be in accordance with table I.

TABLE I. Examination schedule.

First article	Quality conformance	Defects
X	X	101. Dimensions not as specified.
X	-	102. Materials not as specified.
X	-	103. Materials are not corrosion resistant or treated to be corrosion resistant to the applicable storage and operating environment.
X	-	104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
X	-	105. Contractor does not have documentation available for identification of material, material finishes, or treatment.
X	-	106. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the laboratory.
X	X	107. Fire extinguisher missing.
X	X	108. Fungus-moisture proof treatment not as specified.
X	X	109. Treatment and painting not as specified.
X	X	110. Camouflage patterns not as specified.
X	-	111. Wiring and electrical supplies not in compliance with requirements.
X	X	112. Wiring protection not as specified.
X	-	113. Wire identification missing.
X	X	114. Location of equipment in laboratory not as specified in technical manual.
X	X	115. All equipment, supplies, chemicals, binders, books, and manuals not secured within the laboratory to permit air transportability.
X	-	116. Safety not as specified.
X	X	117. Workmanship not as specified.
X	X	118. Metal fabrication not as specified.
X	X	119. Welding not as specified.
X	-	120. Welders certification not made available as specified.
X	X	121. Brazing not as specified.
X	X	122. Bolted connections not as specified.
X	X	123. Riveted connections not as specified.
X	X	124. Machine work not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter and at climatic conditions existing at the place of test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:31Z
Check the source to verify that this is the current version before use.

4.5.2.2 Automatic gas alarm. The automatic gas alarm shall be tested for operation as follows:

a. Energize the system, set the voltage, and test in accordance with manufacturer's instructions.
b. Envelop the detector with a plastic bag (approximately 8 inches square).
c. Using a 14-pound lecture bottle of propane, insert the nozzle into the bag opening and momentarily open the valve to allow a surge of gas into the bag.
d. Seal the bag around the detector by hand and allow 30 seconds for the alarm to activate.
e. Failure of the alarm to activate within 30 seconds shall constitute failure of this test.

4.5.2.3 Water system. The water system shall be examined for leaks when tested as follows:

a. Close gate valve to inlet of on board water tank and open gate valve after pressure regulator. Connect water system to an outside water supply, open faucet and purge system of all air. Close faucet and maintain an inlet water pressure of 45 psig for 30 minutes.
b. Open gate valve to inlet of on board water tank and fill tank from an outside water supply. Close gate valves to inlet of water tank and after pressure regulator. Open faucet and operate on board water pump to purge system of all air. Shut faucet and observe shut off of pump by pressure actuated switch.

The testing and examination shall be conducted for a total of two times. After the first test, the pressure shall be allowed to return to ambient before pressurizing for the second test. Any evidence of leakage shall constitute failure of this test.

4.5.2.4 Air-vacuum system. The air-vacuum system shall be examined for leaks when pressure tested at 30 psig for not less than 20 minutes. Soap solution shall be used at joints, fittings, and connections to aid inspection. The testing and examination shall be conducted for a total of two times. After the first test the pressure shall be allowed to return to ambient before pressurizing for the second test. Any evidence of leakage shall constitute failure of this test.

4.5.2.5 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum: noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 8 equal (horizontal) arc increments, one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:31Z
Check the source to verify that this is the current version before use.

4.5 Inspection schedule.

4.5.1 Examination. Examination shall be in accordance with table I.

TABLE I. Examination schedule.

First article	Quality conformance	Defects
X	X	101. Dimensions not as specified.
X	-	102. Materials not as specified.
X	-	103. Materials are not corrosion resistant or treated to be corrosion resistant to the applicable storage and operating environment.
X	-	104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
X	-	105. Contractor does not have documentation available for identification of material, material finishes, or treatment.
X	-	106. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the laboratory.
X	X	107. Fire extinguisher missing.
X	X	108. Fungus-moisture proof treatment not as specified.
X	X	109. Treatment and painting not as specified.
X	X	110. Camouflage patterns not as specified.
X	-	111. Wiring and electrical supplies not in compliance with requirements.
X	X	112. Wiring protection not as specified.
X	-	113. Wire identification missing.
X	X	114. Location of equipment in laboratory not as specified in technical manual.
X	X	115. All equipment, supplies, chemicals, binders, books, and manuals not secured within the laboratory to permit air transportability.
X	-	116. Safety not as specified.
X	X	117. Workmanship not as specified.
X	X	118. Metal fabrication not as specified.
X	X	119. Welding not as specified.
X	-	120. Welders certification not made available as specified.
X	X	121. Brazing not as specified.
X	X	122. Bolted connections not as specified.
X	X	123. Riveted connections not as specified.
X	X	124. Machine work not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter and at climatic conditions existing at the place of test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:31Z
Check the source to verify that this is the current version before use.

4.5.2.2 Automatic gas alarm. The automatic gas alarm shall be tested for operation as follows:

a. Energize the system, set the voltage, and test in accordance with manufacturer's instructions.
b. Envelop the detector with a plastic bag (approximately 8 inches square).
c. Using a 14-pound lecture bottle of propane, insert the nozzle into the bag opening and momentarily open the valve to allow a surge of gas into the bag.
d. Seal the bag around the detector by hand and allow 30 seconds for the alarm to activate.
e. Failure of the alarm to activate within 30 seconds shall constitute failure of this test.

4.5.2.3 Water system. The water system shall be examined for leaks when tested as follows:

a. Close gate valve to inlet of on board water tank and open gate valve after pressure regulator. Connect water system to an outside water supply, open faucet and purge system of all air. Close faucet and maintain an inlet water pressure of 45 psig for 30 minutes.
b. Open gate valve to inlet of on board water tank and fill tank from an outside water supply. Close gate valves to inlet of water tank and after pressure regulator. Open faucet and operate on board water pump to purge system of all air. Shut faucet and observe shut off of pump by pressure actuated switch.

The testing and examination shall be conducted for a total of two times. After the first test, the pressure shall be allowed to return to ambient before pressurizing for the second test. Any evidence of leakage shall constitute failure of this test.

4.5.2.4 Air-vacuum system. The air-vacuum system shall be examined for leaks when pressure tested at 30 psig for not less than 20 minutes. Soap solution shall be used at joints, fittings, and connections to aid inspection. The testing and examination shall be conducted for a total of two times. After the first test the pressure shall be allowed to return to ambient before pressurizing for the second test. Any evidence of leakage shall constitute failure of this test.

4.5.2.5 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum: noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 8 equal (horizontal) arc increments, one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:31Z
Check the source to verify that this is the current version before use.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Examination. Each pack shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

124. Components not secured in their designated locations for level A, B, or commercial.
125. Materials and containers not as specified for level A or B.
126. Blocking, bracing, and anchoring not in accordance with the referenced documents for level A.
127. Strapping not applied as specified for level B.
128. Tamper-proof seal not applied as specified for commercial.
129. Marking missing, illegible, incorrect, or incomplete for level A, B, or commercial.

5. PACKAGING

5.1 Preservation. (see 6.5)

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. The closed laboratory, with components secured in their designated locations, shall be packed in a close-fitting crate conforming to MIL-C-3774, type I, style A. Blocking, bracing, and anchoring of the laboratory within the crate shall be in accordance with the crate specification and MIL-STD-1186.

5.2.2 Level B. The closed laboratory, with components secured in their designated locations, shall be banded around its circumference, across the door, with two flat steel straps, size 0.035 inch by 1.250 inch, conforming to QQ-S-781, or type I or IV, finish A.

5.2.3 Commercial. The closed laboratory, with components secured in their designated locations, shall not require additional packing, except the door latch shall be protected with a tamper-proof seal, applied in such a manner that the door cannot be opened without destroying the seal.

5.3 Marking.

5.3.1 Military. Marking for military shipment (level A or B) shall be in accordance with MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:31Z
Check the source to verify that this is the current version before use.

5.3.2 Commercial. Commercial marking shall be in accordance with ASTM D 3951 and in addition shall indicate the weight and cube.

6. NOTES

6.1 Intended use. The laboratory is intended for use in the field for the evaluation of aviation fuel, and is designed to be operable at ambient temperatures of -40° F to +125° F and withstand storage temperatures of -50° F to +145° F.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
c. Time frame required for submission of first article (see 3.2).
d. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
e. When the fire extinguishers will not be furnished by the Government (see 3.4).
f. When overcoat in accordance with camouflage patterns is not required (see 3.9).
g. When Government-furnished property is required (see 3.14).
h. When the welding process shall not be the option of the contractor (see 3.18.2).
i. Degree of packing required (see 5.2).

6.3 Government-furnished property. Unless otherwise specified (see 6.2), the contracting officer should arrange to furnish the property specified in 3.4, and, when specified (see 6.2), the property specified in 3.14.

6.4 First article. When a first article inspection is required, the item should be an initial production unit. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the documents' first articles.

6.5 Preservation. There are no preservation requirements applicable to this laboratory. The closed laboratory with all components cushioned and secured in their designated locations, by the means provided, meets the minimum requirements for levels A, B, or commercial preservation. For purposes of preservation/packing level marking the preservation shall be marked level A.

6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and suppliers maintenance and operation manual to be furnished with each laboratory.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:31Z
Check the source to verify that this is the current version before use.

6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of laboratories. A suggested paragraph is as follows:

"Shipment of laboratories shall include repair parts, maintenance tools, operational instructions, and accessories, unless exceptions are provided elsewhere in the contract."

6.8 Subject term (key word) listing.

Aviation fuel
Laboratory
Laboratory, air mobile
Petroleum testing
Quality surveilance

6.9 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 6640-A084

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:31Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements All users of military standardization documents are invited to provide suggestions This form may be detached, folded along the lines indicated, taped along the open edge *(DO NOT STAPLE)*, and mailed In block 5, be as specific as possible about particular problem areas such as wording which required interprotation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems Enter in block 6 any remarks not related to a specific paragraph of the document If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12062 WASHINGTON D C
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

DEPARTMENT OF THE ARMY
U.S. Army Belvoir Research,
Development and Engineering Center
ATTN: STRBE-TSE
Fort Belvoir, VA 22060-5606

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:31Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-52733A(ME)	Laboratory, Airmobile, Aviation Fuel

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:31Z
Check the source to verify that this is the current version before use.